# Companion

Employee journey companion | Solution note

## Purpose and scope

Companion gives an employee one place to understand their onboarding or internal move, ask policy questions and route a blocker to the right named human. It combines journey events, open requests and expected milestones so a missing step becomes visible before a complaint. The prototype uses synthetic data only; it has no connection to live Emirates systems and makes no performance or disciplinary decisions.

## Working employee experience

The authenticated employee sees their own milestones, deadlines, owners and next steps. A deterministic journey engine compares timestamped events with map-defined targets and prerequisites. It identifies overdue steps and conflicting records, such as a completed request without its completion event. Dependencies prevent premature downstream fulfilment. Handoffs reuse linked requests rather than creating duplicates.

Policy questions use local retrieval and cite the source document, version and section. Employees can open the original text. Unknown or unsupported questions abstain and direct the employee toward human clarification. The current runnable demonstration uses rules-based guidance. An optional loopback-only Ollama adapter is implemented, with no external fallback or model action tools. Model answers must be verbatim excerpts present in the cited supplied passages. Actual on-device inference and model quality remain unverified because installation was deferred; a citation does not establish policy applicability.

## Human ownership and friction response

Event delays, request evidence and consented check-ins inform support. There is no inferred employee risk score. The employee reviews a named recipient and chooses separately whether saved feedback accompanies a handoff. The local server builds and persists the case using authoritative journey records. Only its assigned owner can acknowledge it, request information or resolve it with a required outcome note. Resolution records the owner and timestamps, updates the linked request and adds the completion event after prerequisite checks. These are functioning local mock-service queues; they do not contact real HR, IT or facilities services.

## Data ingestion and reuse

The importer supports employees.csv, journey_events.csv, service_requests.csv, pulse_feedback.csv, a policies/ directory of UTF-8 Markdown/text, and structured journey_maps.md tables. Validation checks required inputs, identifiers, dates, joins, owners, policies and dependency cycles before atomic activation. Invalid packs preserve the active dataset. Document provenance and consent filtering are retained. The organiser pack was unavailable at submission preparation. The included example has 12 invented employees and two invented policies; final-pack compatibility remains unverified. Target times use calendar-based rules, not business-day calendars.

## Consent and employee privacy

Pulse records require explicit consent. Missing consent excludes feedback; an explicit refusal takes precedence. Saving a check-in and attaching it to a case are distinct choices. Employees can export their own normalized records. Withdrawal deletes saved feedback, structured case attachments and the employee's persisted conversation; future owner reads exclude feedback. It cannot undo information already read or manually copied into other free text. Original input files are not modified, and reimport re-evaluates source consent. The employee interface stores no employee records or credentials in localStorage.

## Access controls and oversight

Local accounts are provisioned for employees, named owners, managers and an administrator. Server-side identity determines access; request-supplied identities cannot select another employee. Passwords use salted scrypt hashes. Opaque sessions use HttpOnly, SameSite cookies, expire after 30 minutes and support logout; write operations require same-origin requests and a session token. Data persists in an ignored local store with restrictive file permissions. The server binds to loopback. There are no external AI, analytics or font dependencies configured. This is prototype authorization, not enterprise SSO or a security certification; machine-wide network isolation has not been verified.

## Nice-to-haves and validation

English and Arabic controls include RTL layout and a limited local Arabic retrieval vocabulary. Imported labels and policy excerpts remain in their source language; full bilingual quality is not claimed. Managers see only assigned onboarding milestones, excluding conversations, feedback and scores. The administrator sees aggregate overdue-step counts with groups below five suppressed. The fresh example has two reportable groups of six. Fifty-four automated tests pass, covering journey rules, import validation, local inference boundaries, access isolation, named-owner resolution and persistent consent deletion. A browser walkthrough verified employee handoff, owner resolution, journey updates and Arabic layout.

## Pilot path and submission

The runnable repository includes setup steps and a complete example pack. Run npm run setup, then npm start; no app dependencies or model downloads are required. Before a pilot, validate the launch pack and hidden-style cases, integrate approved enterprise identity and service adapters, establish retention and recovery procedures, and verify local-model quality and machine egress. Confirm owner coverage and business-day targets with service teams. Reuse relies on configurable maps, policies and ownership rather than employee-specific rules. The demo video is included at demo-video/demo.mp4; rehearsal scripts and private runtime state are excluded from the repository.